The Riemann Integral

Bartle & Sherbert §7.1 (7.1.1–7.1.7), §7.2 (7.2.1–7.2.13), §7.3 (7.3.1–7.3.18) · FYUGP pattern (April 2026), doubled

HOW THIS PAPER WORKS

A real FYUGP paper has 10 short (3 marks, ceiling 24), 8 paragraph (6 marks, ceiling 36) and 2 essay (10 marks, answer one) questions. This paper has **twice as many** (20 + 16 + 4 = 40 questions, 196 marks), all from Module 3. Each of the module's 32 atomic concepts (T01–T32) and 8 core application types is the main point of exactly one question.

Time	4 hours (or two sittings of 2 hours)
Maximum	140 marks (A ceiling 48, B ceiling 72, C answer any two = 20)

PYQ MTS-25 Q6 = MTS 6B 10, March 2025, question 6 (old CBCSS course, same textbook). "also" = another paper asked the same thing. "adapted" = a 1-mark MCQ turned into a 3-mark or 6-mark question with full justification.

Section A · Short answer

All questions can be answered. Each carries 3 marks. Ceiling 48.

1. Define a partition of [a,b] and the norm of a partition. Find the norms of $P_1 = (0, 1, 2, 4)$, $P_2 = (0, 2, 3, 4)$, and $P_3 = (0, 1, 1.5, 2, 3.4, 4)$ on [0,4]. Which of $Q_1 = (2, 4, 6, 8, 9)$, $Q_2 = (2, 6, 9)$, $Q_3 = (2, 5, 7, 9)$ on [2,9] has the largest norm?

PYQ MTS-26 Q4 **also MAT-22 Q3, MAT-21 Q5** **MAT-22 MCQ 16 · adapted**

---

2. Define a tagged partition and the Riemann sum $S(f; \dot{P})$ of a function f : $[a, b] \to \mathbb{R}$. For $f(x) = x^2$ on [0,4] and P = (0,1,2,4), compute $S(f; \dot{P})$ with tags at left endpoints and at right endpoints.

PYQ MTS-22 Q4

---

3. Let f : $[a, b] \to \mathbb{R}$. State the ε–δ_ε definition of "f is Riemann integrable on [a,b]" (including the notation $\mathcal{R}[a, b]$ and $\int_a^b f$). Why is $\int_a^b f$ not an ordinary function limit as $\|P\| \to 0$?

PYQ MAT-21 MCQ 9 · adapted

---

4. Prove that if $f \in \mathcal{R}[a, b]$, then the value of the integral $\int_a^b f$ is uniquely determined.

---

5. Prove directly from Definition 7.1.1 that every constant function f(x) = k on [a,b] is Riemann integrable on [a,b], and find $\int_a^b k$.

PYQ MTS-25 Q4

---

6. If $g \in \mathcal{R}[a,b]$ and f(x) = g(x) except for a finite number of points in [a,b], prove that $f \in \mathcal{R}[a,b]$ and $\int_a^b f = \int_a^b g$ (proving the case of one exceptional point suffices). Hence find $\int_0^2 h$ where h(x) = 2 on [0,1), h(1) = 3, and h(x) = 1 on (1,2]. What happens if f(x) = 0 except at finitely many points, or if f(x) = g(x) off the infinite set $\{1/n : n \in \mathbb{N}\}$?

PYQ MAT-22 MCQ 18 · adapted

also MAT-21 MCQ 13, MTS-22 MCQ 7 · adapted

---

7. State and prove the Boundedness Theorem for Riemann integrable functions. Deduce that $H(1/k) = k (k \in \mathbb{N}), H(x) = 0$ elsewhere on [0,1], is not in $\mathcal{R}[0,1]$. Does $f \in \mathcal{R}[a,b]$ imply f is continuous or monotone? Is every bounded function integrable?

PYQ MAT-22 MCQ 15 · adapted

also MAT-21 MCQ 11, 17, MTS-22 MCQ 8, 9 · adapted

---

8. If J is a subinterval of [a,b] with endpoints c < d and $\varphi_J(x) = 1$ on J, 0 elsewhere on [a,b], prove that $\varphi_J \in \mathcal{R}[a,b]$ and $\int_a^b \varphi_J = d - c$. Hence evaluate $\int_0^4 g$ where g(x) = 3 for $0 \le x \le 2$ and g(x) = 2 for $2 < x \le 4$.

PYQ MAT-21 MCQ 10 · adapted

---

9. For $f \in \mathcal{R}[a,b]$ and $\alpha < \beta$ in [a,b], define $\int_\beta^\alpha f$ and $\int_\alpha^\alpha f$. State the additivity formula for arbitrary $\alpha, \beta, \gamma \in [a,b]$ and verify it for f(x) = x with $\alpha = 3, \beta = 1, \gamma = 0$.

10. Explain why the First Form of the Fundamental Theorem of Calculus (Theorem 7.3.1) cannot be applied to $H(x) = 2\sqrt{x}$ on [0,b], nor to $K(x) = x^2 \cos(1/x^2) (x \in (0,1]), K(0) = 0$, on [0,1], even though H is continuous on [0,b] and K is differentiable at every point of [0,1].

---

11. Define the indefinite integral F of $f \in \mathcal{R}[a,b]$ with basepoint a. Prove that if $|f(x)| \leq M$ on [a,b], then $|F(z)F(w)| \leq M|z-w|$ for all $z, w \in [a,b]$, and deduce that F is continuous on [a,b].

PYQ MAT-21 Q6

---

12. Find the indefinite integral $F(x) = \int_{-1}^{x} \operatorname{sgn}$ of $f(x) = \operatorname{sgn} x$ on [-1,1] with basepoint -1. Is F an antiderivative of f on [-1,1]? What about $H(x) = \int_0^x h$ for Thomae's function h on [0,1]?

---

13. Define a null set in $\mathbb{R}$. Prove that the set $\mathbb{Q} \cap [0,1]$ of rational numbers in [0,1] is a null set. State whether subsets and unions of null sets are null, and whether every null set is countable.

PYQ MTS-25 Q7 **also MTS-26 Q7**
MTS-22 MCQ 10 · adapted

---

14. Using the First Form of the Fundamental Theorem of Calculus: (a) state $\int_a^b f(x)\,dx$ when f is continuous and F is an antiderivative of f on [a,b]; (b) evaluate $\int_{-2}^{3} g(x)\,dx$ where g(x) = x for $|x| \geq 1$ and g(x) = -x for $|x| < 1$, using $G(x) = \frac{1}{2}|x^2 - 1|$; (c) show that $\int_a^b |x|\,dx = B(b) - B(a)$ where $B(x) = -\frac{1}{2}x^2 (x < 0), \frac{1}{2}x^2 (x \geq 0)$, stating the exceptional set E in (b) and (c).

PYQ MAT-22 MCQ 27, 28, 29 · adapted

also MAT-21 MCQ 19, 20 · adapted

---

15. Evaluate the integral $\int_0^1 x^2(1 - \sqrt{x})\,dx$, justifying each hypothesis of the Fundamental Theorem of Calculus that you use.

PYQ MAT-22 Q35

---

16. (a) If f is continuous on [-1,1], use the Substitution Theorem to show that $\int_0^{\pi/2} f(\cos x)\,dx = \int_0^{\pi/2} f(\sin x)\,dx = \frac{1}{2}\int_0^{\pi} f(\sin x)\,dx (and$ is not in general equal to $\int_0^{\pi} f(\sin x)\,dx).(b)$ If $f \in \mathcal{R}[a,b]$ and g(y) := f(y - c) on [a+c,b+c], show that $\int_{a+c}^{b+c} g = \int_a^b f$.

PYQ MAT-22 MCQ 19, 26 · adapted

---

17. If f : $[0,1] \to \mathbb{R}$ is continuous and $\int_0^x f = \int_x^1 f$ for all $x \in [0,1]$, prove that $f \in \mathcal{R}[0,1]$ and f(x) = 0 for all $x \in [0,1]$.

PYQ MAT-22 MCQ 30 · adapted

18. Let f be continuous on [a,b] with $f(x) \geq 0$ for all $x \in [a,b]$. If $\int_a^b f = 0$, prove that f(x) = 0 for all $x \in [a,b]$. Show by an example that the continuity of f cannot be dropped.

PYQ MAT-22 MCQ 23 · adapted

---

19. State the rule for differentiating $G(x) = \int_{u(x)}^{v(x)} f(t)\,dt$ when f is continuous and u, v are differentiable. Hence find F'(x) on [0,1] for: $(a) F(x) = \int_0^{x^2} (1+t^3)^{-1}\,dt$; $(b) F(x) = \int_{x^2}^{x} \sqrt{1+t^2}\,dt$.

PYQ MTS-22 Q14

---

20. (Mean Value Theorem for Integrals) If f : $[a,b] \to \mathbb{R}$ is continuous on [a,b] (a < b), prove that there exists $c \in [a,b]$ such that $\int_a^b f = f(c)(b-a)$. Does this hold for step functions?

Section B · Paragraph / problem

All questions can be answered. Each carries 6 marks. Ceiling 72.

21. Suppose f and g are in $\mathcal{R}[a,b]$. Prove that: $(a) f+g \in \mathcal{R}[a,b]$ and $\int_a^b (f+g) = \int_a^b f + \int_a^b g$; (b) if $f(x) \leq g(x)$ for all $x \in [a,b]$, then $\int_a^b f \leq \int_a^b g (and$ show why strict inequality < need not hold).

PYQ MAT-21 Q29 **also MTS-22 Q5, MTS-25 Q5**
MAT-22 MCQ 14 · adapted

---

22. Define Thomae's function h on [0,1]. Prove directly from Definition 7.1.1 that $h \in \mathcal{R}[0,1]$ and $\int_0^1 h = 0$. Explicitly list all elements of the set $E = \{x \in [0,1] : h(x) \geq 1/5\}$.

---

23. State and prove the Cauchy Criterion for the Riemann integrability of f : $[a,b] \to \mathbb{R}$.

PYQ MAT-21 Q8

---

24. Prove using the Cauchy Criterion that Dirichlet's function (f(x) = 1 on $\mathbb{Q} \cap [0,1]$, 0 on $[0,1] \setminus \mathbb{Q}$) is not Riemann integrable on [0,1]. Also prove that h(x) = x + 1 on $\mathbb{Q} \cap [0,1]$, $h(x) = 0$ on $[0,1] \setminus \mathbb{Q}$, is not in $\mathcal{R}[0,1]$.

PYQ MAT-22 Q4
also MAT-22 MCQ 17, MAT-21 MCQ 12, 16 · adapted

---

25. State and prove the Squeeze Theorem for Riemann integrals. Use it with step functions on an n-part equal partition to prove that h(x) = x belongs to $\mathcal{R}[0,1]$ with $\int_0^1 x\,dx = \frac{1}{2}$.

PYQ MTS-26 Q18 **also MTS-22 Q6, MAT-21 Q7**

26. Prove that every step function $\varphi : [a, b] \to \mathbb{R}$ is Riemann integrable, and derive the formula $\int_a^b \varphi = \sum_{j=1}^m k_j(d_j - c_j)$. Use it to evaluate $\int_0^3 g$ where g(x) = 2 on [0,1] and g(x) = 3 on (1,3].

PYQ MTS-26 Q5 **also MAT-22 Q16**

---

27. If $f \in \mathcal{R}[a, b]$ and $[c, d] \subseteq [a, b]$, prove that the restriction of f to [c,d] belongs to $\mathcal{R}[c, d]$. Deduce by mathematical induction that if $a = c_0 < c_1 < \cdots < c_m = b$, then $\int_a^b f = \sum_{i=1}^m \int_{c_{i-1}}^{c_i} f$.

PYQ MAT-22 Q37b

---

28. Let f be bounded by M on [a,b] and suppose the restriction of f to [c,b] belongs to $\mathcal{R}[c, b]$ for every $c \in (a, b)$. Prove that $f \in \mathcal{R}[a, b]$ and $\lim_{c \to a^+} \int_c^b f = \int_a^b f$. Deduce that $g(x) = \sin(1/x)(x \in (0, 1]), g(0) = 0$, belongs to $\mathcal{R}[0, 1]$, and show by an example that the boundedness of f cannot be dropped.

PYQ MAT-22 MCQ 24, 25 · adapted
also MAT-21 MCQ 18 · adapted

---

29. State and prove the First Form of the Fundamental Theorem of Calculus (prove the case $E = \{a, b\}$ in full and show how the general finite set E follows).

PYQ MTS-26 Q8

---

30. State and prove the Substitution Theorem for Riemann integrals. Use it to evaluate $\int_1^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt$, $\int_1^4 \frac{\cos\sqrt{t}}{\sqrt{t}}\,dt$, $\int_0^1 t\sqrt{1+t^2}\,dt$, $\int_0^2 t^2(1+t^3)^{-1/2}\,dt$, and $\int_1^4 \frac{\sqrt{1+\sqrt{t}}}{\sqrt{t}}\,dt$. Finally, explain why the Substitution Theorem with $\varphi(t)=\sqrt{t}$ cannot be applied to $\int_0^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt$, and evaluate $\int_0^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt$ using the First Form of the Fundamental Theorem.

PYQ MTS-25 Q6 **also MAT-22 Q18, MAT-21 Q19**

---

31. State Lebesgue's Integrability Criterion. Use it to determine the Riemann integrability on [0,1] of: (a) any step function; (b) any monotone function; $(c) G(1/n)=1/n (n\in\mathbb{N}), G(x)=0$ elsewhere; (d) Dirichlet's function; (e) Thomae's function.

PYQ MAT-22 Q6

---

32. State and prove the Composition Theorem for Riemann integrable functions. Show by an explicit counterexample that the continuity of the outer function φ cannot be dropped.

PYQ MAT-21 Q4

---

33. If $f\in\mathcal{R}[a,b]$ with $|f(x)|\le M$ on [a,b], prove that $|f|\in\mathcal{R}[a,b]$ and $|\int_a^b f|\le\int_a^b|f|\le M(b-a)$. Show by an example that $|f|\in\mathcal{R}[a,b]$ does not imply $f\in\mathcal{R}[a,b]$.

PYQ MTS-25 Q8 **also MTS-22 Q7, MAT-21 Q20**

---

34. State and prove the Product Theorem: if $f, g \in \mathcal{R}[a,b]$, then fg $\in \mathcal{R}[a,b]$. Give an example to show that $\int_a^b$ fg $\neq (\int_a^b f)(\int_a^b g)$ in general.

PYQ MTS-26 Q6 **also MAT-22 Q17, MAT-21 Q18**

---

35. State and prove the Integration by Parts Theorem for Riemann integrals. Use it to evaluate $\int_0^1 xe^x\,dx$.

---

36. State and prove Taylor's Theorem with the Integral Form of the Remainder. Verify the theorem for $f(x) = e^x$ on [0,1] (a = 0, b = 1) with n = 1 and n = 2.

PYQ MTS-25 Q19 **also MAT-22 Q32**

Section C · Essay

Answer any two. Each carries 10 marks.

37. Prove that if f : $[a, b] \to \mathbb{R}$ is continuous on [a,b], then $f \in \mathcal{R}[a, b]$. State clearly where each hypothesis is used in the proof. Is every function in $\mathcal{R}[a, b]$ continuous on [a,b]? Is every function in $\mathcal{R}[a, b]$ uniformly continuous on [a,b]? Justify with examples.

PYQ MTS-25 Q18 **also MAT-21 Q30, MTS-22 Q19**
MAT-22 MCQ 21, MAT-21 MCQ 15 · adapted

---

38. Prove that if f : $[a, b] \to \mathbb{R}$ is monotone on [a,b], then $f \in \mathcal{R}[a, b]$. Illustrate the proof by computing the lower and upper step-function integrals $\int_0^1 \alpha$ and $\int_0^1 \omega$ for $f(x) = x^2$ on [0,1] with n = 4 equal subintervals. Is every function in $\mathcal{R}[a, b]$ monotone on [a,b]?

PYQ MTS-26 Q19 **also MAT-22 Q30**
MAT-22 MCQ 22 · adapted

---

39. State and prove the Additivity Theorem: let f : $[a, b] \to \mathbb{R}$ and $c \in (a, b)$; prove that $f \in \mathcal{R}[a, b]$ if and only if its restrictions to [a,c] and [c,b] are both in $\mathcal{R}$, and in this case $\int_a^b f = \int_a^c f + \int_c^b f$.

PYQ MTS-26 Q25 **also MAT-22 Q37a, MTS-22 Q25**

---

40. State and prove the Second Form of the Fundamental Theorem of Calculus (both the pointwise form at $c \in [a, b]$ and the global consequence when f is continuous on [a,b]). Give an example of a function $f \in \mathcal{R}[-1, 1]$ whose indefinite integral fails to be differentiable at 0.

PYQ MTS-25 Q25 **also MAT-21 Q37a, MAT-22 Q33**

Coverage

TOPIC	QUESTIONS
T01 Partitions, subintervals, norm	1
T02 Tagged partitions, Riemann sums	2
T03 Definition of $\mathcal{R}[a,b]$ (7.1.1)	3
T04 Uniqueness of the integral (7.1.2)	4
T05 Integrals from the definition (7.1.4)	5
T06 Finite exceptional set (7.1.3)	6
T07 Linearity and monotonicity (7.1.5)	21
T08 Boundedness Theorem (7.1.6)	7
T09 Thomae's function (7.1.7)	22
T10 Cauchy Criterion (7.2.1)	23
T11 Dirichlet and relatives $\notin \mathcal{R}$ (7.2.2b)	24
T12 Squeeze Theorem (7.2.3)	25
T13 Elementary step functions φ_J (7.2.4)	8
T14 Step functions $\in \mathcal{R}[a,b]$ (7.2.5)	26
T15 Continuous $\Rightarrow \mathcal{R}[a,b]$ (7.2.7) & applications	18, 20, 37
T16 Monotone $\Rightarrow \mathcal{R}[a,b]$ (7.2.8)	38
T17 Additivity Theorem (7.2.9)	39
T18 Restriction & subdivision (7.2.10, 7.2.11, Ex 7.2.11)	27, 28
T19 Oriented integrals $\int_\alpha^\beta f$ (7.2.12, 7.2.13)	9
T20 FTC First Form (7.3.1)	14, 15, 29
T21 Examples and limits of FTC 1 (7.3.2)	10, 14, 15
T22 Indefinite integral is Lipschitz (7.3.3, 7.3.4)	11
T23 FTC Second Form (7.3.5, 7.3.6) & variable limits	17, 19, 40

T24 When F is not an antiderivative (7.3.7)	12, 40
T25 Substitution Theorem (7.3.8, 7.3.9)	16, 30
T26 Null sets & $\mathbb{Q} \cap [0,1] (7.3.10, 7.3.11)$	13
T27 Lebesgue's Integrability Criterion (7.3.12, 7.3.13)	31
T28 Composition Theorem (7.3.14)	32
T29 Absolute value $\lvert f \rvert \in \mathcal{R}[a,b] (7.3.15)$	33
T30 Product Theorem fg $\in \mathcal{R}[a,b] (7.3.16)$	34
T31 Integration by Parts (7.3.17)	35
T32 Taylor's Theorem with Integral Remainder (7.3.18)	36

All 74 past questions appear (plus the 3 additional MTS-22 descriptive rows in Part 8: 77 past questions in total — 46 written, 31 MCQs).